

Proximity-dependent biotinylation identifies a suite of candidate effector proteins from *Fusarium graminearum*

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SUMMARY

Fusarium graminearum is a fungal pathogen that causes Fusarium head blight in cereal crops. The identification of proteins secreted from pathogens to overcome plant defenses and cause disease, collectively known as effectors, can reveal the etiology of a disease process. Proximity-dependent biotin identification (BioID) was used to identify potential effector proteins secreted *in planta* by *F. graminearum* during the infection of *Arabidopsis*. Mass spectrometry analysis of streptavidin affinity-purified proteins revealed over 300 proteins from *F. graminearum*, of which 62 were candidate effector proteins (CEPs). An independent analysis of secreted proteins from axenic cultures of *F. graminearum* showed a 42% overlap with CEPs, thereby assuring confidence in the BioID methodology. The analysis also revealed that 19 out of 62 CEPs (approx. 30%) had been previously characterized with virulence function in fungi. The functional characterization of additional CEPs was undertaken through deletion analysis by the CRISPR/Cas9 method, and by overexpression into *Triticum aestivum* (wheat) leaves by the *Ustilago hordei* delivery system. Deletion studies of 12 CEPs confirmed the effector function of three previously characterized CEPs and validated the function of another four CEPs on wheat inflorescence or vegetative tissues. Lastly, overexpression in wheat showed that all seven CEPs enhanced resistance against the bacterial pathogen *Pseudomonas syringae* DC3000.

Keywords: BioID, effectors, *Fusarium graminearum*.

INTRODUCTION

Fusarium head blight (FHB) is a devastating disease of *Triticum aestivum* (wheat), *Zea mays* (maize) and other small grain crops and is caused by several related *Fusarium* species. *Fusarium graminearum* can infect seedlings, floral tissues and stems, resulting in reduced grain quality through mycotoxin deposition. To maintain high-quality grain production, research efforts have produced partially resistant cereal crops and have identified over 100 resistance quantitative trait loci in wheat (Buerstmayr et al., 2019). However, the underlying mechanisms involved in partial or complete FHB resistance remain unclear.

Fungal pathogens utilize suites of secreted molecules, called effectors, to sabotage the host cellular machinery to promote infection (Selin et al., 2016). Effectors come in many flavors, including proteins, peptides, small RNAs and metabolites such as mycotoxins (Derbyshire et al., 2019;

Jiang et al., 2020). Our overarching goal is to delineate the mechanisms of how the fungal effectors promote virulence or induce defense responses in the plant. Although great strides have been made in the identification and characterization of proteinaceous effectors in bacteria and oomycetes, a similar statement cannot be made for phytopathogenic fungal effectors. This is mainly because sequence-based features are lacking in fungal genomes and thus relatively broad criteria observed in other pathogens, such as the presence of a secretion signal, small protein size, cysteine residue richness, diversifying selection and genome location, are being used to identify effectors. The majority of fungal effectors are also under constant selective pressure from the host, resulting in higher rates of non-synonymous mutations and a so-called ‘two-speed’ genome (Depotter & Doehlemann, 2020). ‘Two-speed’ refers to areas of the genome that are rich in repetitive sequences and transposable elements, serving as a hotbed for adaptive evolution compared with the core.

Numerous proteome, transcriptome, and bioinformatic studies have been undertaken to identify the effector repertoire of *F. graminearum* (Alouane et al., 2021; Fabre et al., 2019; Pan et al., 2018; Paper et al., 2007; Rampitsch et al., 2013). Each of these methods has its limitations: predominantly they lack *in planta* context, and technically proteins are subjected to extraction and purification conditions that make it impossible to identify proteins that interact transiently during the initial stages of plant–pathogen interactions. As mentioned, predicting effectors based on predetermined effector properties may exclude effectors lacking these properties but possessing effector function (Alouane et al., 2021).

Proximity-dependent biotin identification, also known as BioID, has emerged as a new tool in proteomics to identify protein interactions *in vivo* (Gingras et al., 2019; Kim & Roux, 2016). BioID involves the use of a mutated biotin ligase isolated from *Escherichia coli*, BirA*, to promiscuously biotinylate any proximal proteins within a 10-nm range that contain a primary amine. The mutated ligase has a lower affinity for its substrate (bioAMP) than the non-mutated ligase, allowing it to biotinylate proximal proteins efficiently. Once proteins are biotinylated, they can be extracted under denaturing conditions, isolated using affinity purification and identified by mass spectrometry. This technique can reveal novel protein–protein interactions and, more importantly, allows a ‘real-time’ glimpse of protein interactions over several hours. More broadly, BioID can capture an image of protein populations within specific cellular compartments or of protein complexes under specific environmental conditions (Khan et al., 2021). Thus, the technique allows us to identify potential effectors during the early stages of plant–pathogen interactions.

BioID has been successfully used in plants. BioID has been used in *Oryza sativa* (rice) protoplasts to identify proteins involved in flowering or vegetative growth (Lin et al., 2017). In *Nicotiana benthamiana*, BioID examined plant targets of the bacterial effector AvrPto (Conlan et al., 2018). Several interacting proteins with putative roles in plant immunity were identified. Our group deployed BioID in Arabidopsis to identify targets of the effector HopF2 from *Pseudomonas syringae* (Khan et al., 2018). Transgenic Arabidopsis expressing a cytoplasmic BirA* and a membrane-associated effector HopF2 were used to identify 19 proteins associated with HopF2, revealing both known and novel immunity-related proteins (Khan et al., 2018). In this report, we use the *F. graminearum*–Arabidopsis pathosystem to identify potential effectors that translocate inside plants. Selected candidate effector proteins (CEPs) were subjected to functional analysis through deletion by CRISPR/Cas9 and by overexpression in wheat by the *Ustilago* expression system (Alonso et al., 2020).

RESULTS AND DISCUSSION

Optimization of *F. graminearum* infection in Arabidopsis seedlings

The Arabidopsis–*F. graminearum* 96-well plate infection assay was used in our BioID experiments, with modifications (Schreiber et al., 2011). The modifications included the addition of biotin and dexamethasone (Dex) to the growth medium. To ensure that seedling infection by *F. graminearum* was not altered by the modifications, Arabidopsis seedlings inoculated with *F. graminearum* spores were examined for visual phenotypes associated with infection. Infected Arabidopsis seedlings were grown for 24 h in the presence of biotin and Dex. The samples were collected after 24, 48 and 72 h and stained with Trypan Blue (Koch & Slusarenko, 1990). Visually, the first 24 h did not show any infection but, after 48 h post-infection (48 hpi), some cell death was visible at the tips of the cotyledons (Figure 1a). After 72 hpi, the leaves became chlorotic and dark spots representing dead tissue were seen in cotyledon leaves, indicating the breach of host cells (Figure 1a). As cell death occurred after 72 h of infection, we chose the 60-h time point for tissue collection to maximize infection and limit any contamination of the samples from lysed Arabidopsis and *F. graminearum* cells. At the earlier time points (24–48 hpi), the infection had not progressed enough for *F. graminearum* proteins to be secreted into the plant or the proteins that may be secreted during this early infection stage are in low abundance and are not identified by the BioID technique. However, newer iterations of BirA*, such as TurboID, facilitates shorter labeling time and could overcome some of the drawbacks encountered with the original version (Branon et al., 2018).

The assay system was also used to optimize the quantity, and the incorporation of biotin in the Arabidopsis seedlings with the expression of BirA* in the cytoplasm (cBirA*) and in the membrane (mBirA*) (Figure 1b). Biotinylation levels were saturated at 1 mM, and no additional biotinylated proteins were detected by immunoblot with Streptavidin-HRP conjugate at higher biotin concentrations (Figure 1b). BirA* expression in all of the protein extracts was confirmed by immunoblot with hemagglutinin (HA) antibodies (Figure 1b).

Identification of CEPs

The biotin-labeled proteins were purified by streptavidin affinity chromatography (Figure 2). The affinity for streptavidin was evident, with the enrichment of biotin-labeled proteins in both cytoplasm and membrane fractions, and we observed more bound proteins in the cytoplasmic fraction after infection (compare infected/bound in Figure 2a,b). Endogenous enzymes belonging to the carboxylase family require biotin and co-elute in all the affinity purification steps (Che et al., 2003). Biotinylated proteins from

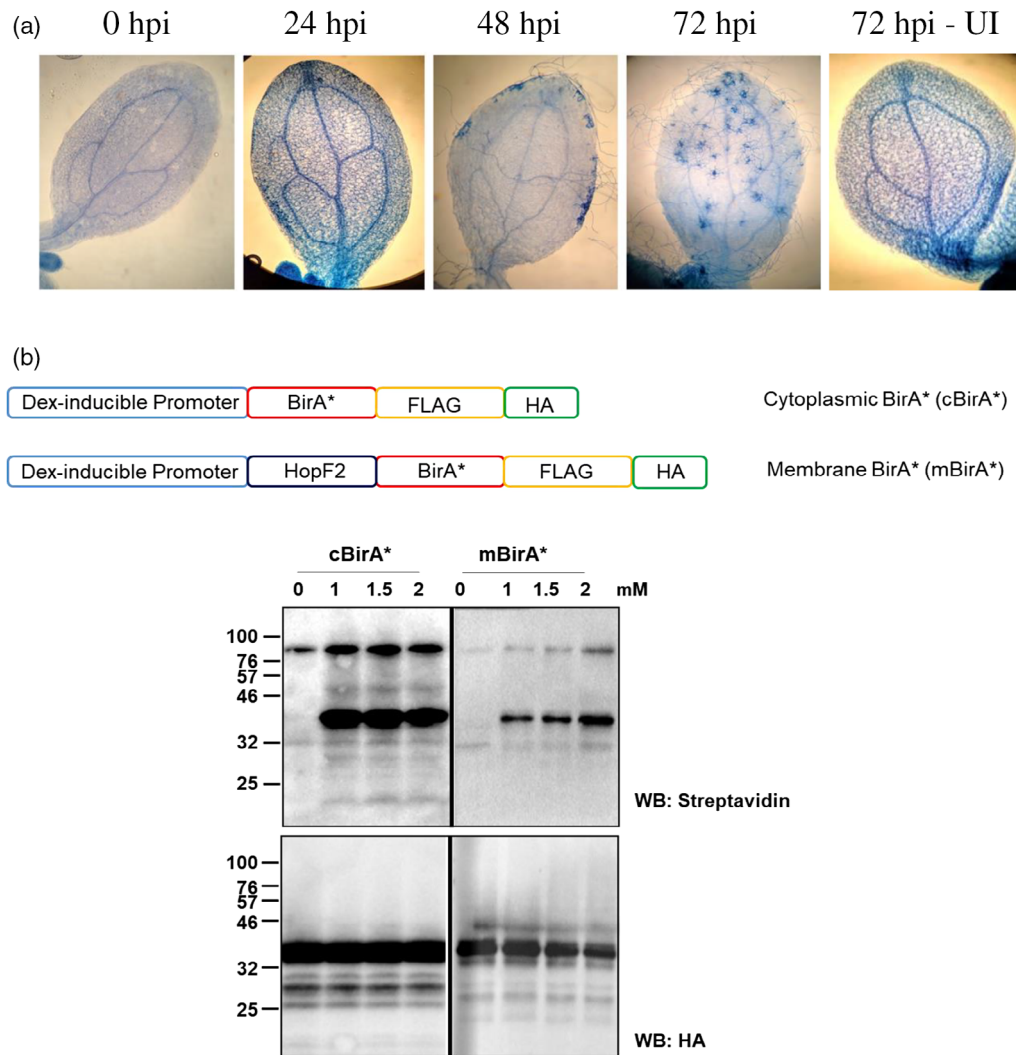


Figure 1. Optimization of infection by *Fusarium graminearum* and biotin uptake in Arabidopsis BirA* lines.

(a) Five-day-old Arabidopsis seedlings were infected with *F. graminearum* spores and grown in the presence of 30 μ M dexamethasone (Dex) and 1 mM biotin. The infected tissues were stained with Trypan blue stain and imaged after 24, 48 and 72 hours post-infection (hpi). For comparison, an uninfected leaf at 72 hpi (UI) is shown. Seedlings were visualized using an SZX16 stereomicroscope (Olympus, <https://www.olympus-global.com>), and images were captured by an Infinity 3 camera and Infinity Analyze and Capture software v6.2.0 (Lumenera, <https://www.lumenera.com>).

(b) Biotin optimization for BiolD. Immunoblot of protein lysates from Arabidopsis seedlings expressing cytoplasmic BirA* (cBirA*) and membrane BirA* (mBirA*) grown in a biotin concentration gradient of 0–2 mM and infected with *F. graminearum* spores for 60 hpi. Biotinylated proteins were detected with streptavidin-HRP conjugate (top panel). Hemagglutinin (HA) antibodies were used to confirm BirA* expression. BirA* expression was constitutive across all biotin concentrations and treatments.

non-infected (Mock) and infected Arabidopsis seedlings isolated by affinity purification were sampled at 60 hpi and analyzed by mass spectrometry. A summary of the experimental set-up with control and experimental treatments is outlined in Table 1.

MASCOT outputs for all treatments and the affinity-purified proteins from mock and infected Arabidopsis seedlings are presented (Datasets 1–5). A total of 351 *Fusarium* proteins were identified as potential effectors (Table S1). However, after considering the criteria that we

established, i.e. the removal of proteins with a Mascot score of < 20, the removal of proteins present in both mock (non-infected) and control (Dex treatment alone) treatments, the removal of proteins present in only one biological replicate (265 proteins removed) and the removal of 24 ribosomal proteins (Table S1), we identified 62 CEPs at the time point of 60 hpi (Table 2). Among the 62 CEPs, 19 and 13 were identified in the cBir* and mBir* samples, respectively, and the rest of the 30 proteins were identified in both the cBir* and mBir* samples (Table 2).

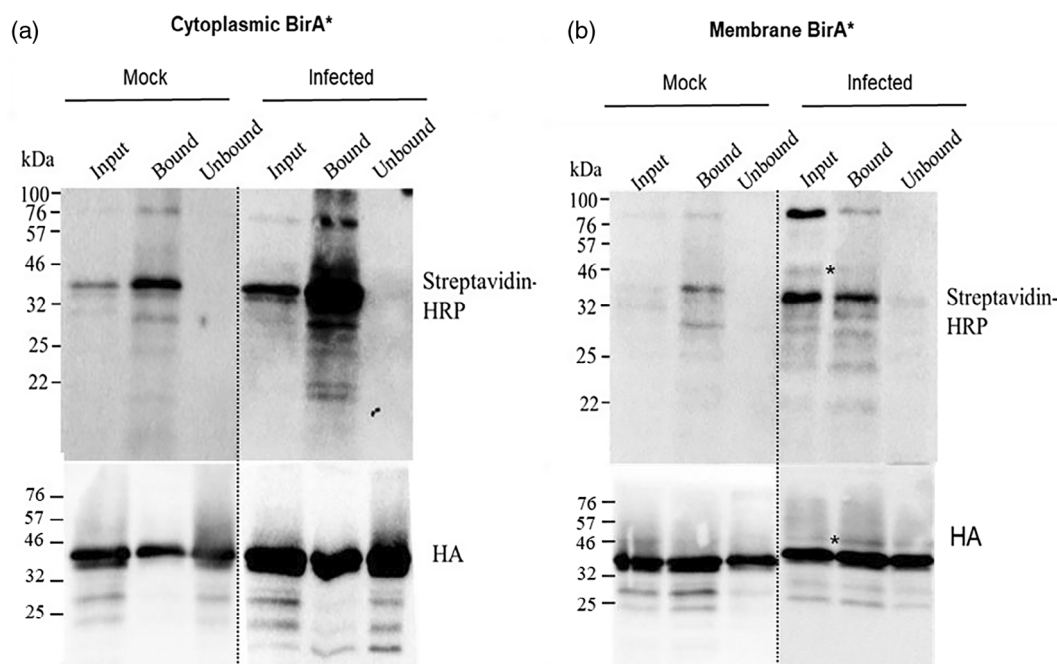


Figure 2. Affinity purification of candidate effector proteins (CEPs) from Arabidopsis plants infected with *Fusarium graminearum*. Biotinylated proteins containing CEPs were isolated at 60 hpi from Arabidopsis seedlings either not infected (Mock) or infected with *F. graminearum* spores (Infected) expressing: (a) cytoplasmic BirA* and (b) membrane BirA*, grown in 30 μ M dexamethasone and 1 mM biotin. Biotinylated proteins were detected in total protein extract (input) and at higher levels in purified biotinylated proteins (bound) by the streptavidin-HRP conjugate. No biotinylated proteins were detected in the affinity purification supernatant fraction (unbound). BirA* expression was confirmed by immunoblot analysis with hemagglutinin (HA) antibodies for all fractions. Dotted lines indicate the demarcation of cropped figures.

Table 1 Outline of the BioID experiments for mass spectrometry analysis

Data set ^a	Treatment ^b	Infection status ^c	Dexamethasone
Dataset 1	cBirA*	mock + infected	–
Dataset 2	cBirA*	mock + infected	+
Dataset 3	mBirA*	mock + infected	–
Dataset 4	mBirA*	mock + infected	+
Dataset 5	Growth media/Col-0	mock + infected	+

^aData sets are included in the Supporting Information.

^bThe Arabidopsis plants express BirA* in the cytoplasm (cBirA*) and the membrane (mBirA*).

^cInfected, inoculated with *Fusarium graminearum* spores; Mock, not inoculated.

Arabidopsis proteome associated with *Fusarium* infection

Previously, BirA* was used to identify both soluble and membrane-associated proteomes in Arabidopsis (Khan et al., 2018). In this study, Arabidopsis transgenic plants expressing BirA* were used to identify Arabidopsis proteins following infection with *F. graminearum*, and, after 60 hpi, a total of 59 proteins were enriched in both cytoplasmic and membrane compartments (Table S2). Gene ontology analysis showed that most of the proteins were significantly enriched for their role in small molecule metabolism (75%;

2.98E-03), such as purine or pyrimidine biosynthesis, and cellular processes (25%; 5.05E-03), such as the establishment/maintenance of cell polarity and cytoskeleton organization. It is envisaged that many of them are involved in plant defense; however, we limited this study to the role of CEPs in virulence and plant immunity.

Validation of CEPs

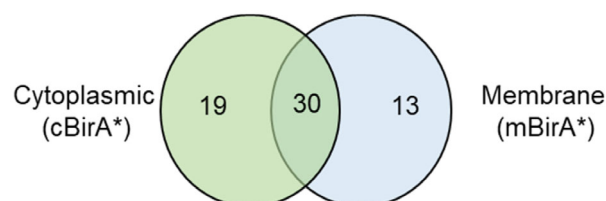
To validate the data obtained from the BioID experiments, we performed an independent experiment and analyzed secreted proteins from the medium of axenic cultures of *F. graminearum* (Table S3). The spores were grown in nutrient-poor conditions with wheat heads, conducive to the expression of effectors such as the mycotoxin deoxynivalenol (DON). We identified 129 proteins, and a comparison between the secreted proteins and the CEPs showed an approximately 42% overlap, with the majority enriched in both the cytoplasm and the membrane fractions (Table S3). We ranked the CEPs based on their presence in five data sets: the secretome from this study; three published secretomes; and one transcriptome study (Table S4) (Fabre et al., 2019; Pan et al., 2018; Paper et al., 2007; Ram-pitsch et al., 2013). The transcriptome data set is particularly relevant, as it was obtained from tissues that had yet to display any disease phenotype and, therefore, genes expressed in asymptomatic tissues are a good indicator of

Table 2 Identification of candidate effector proteins from both cBirA*- and mBirA*-bound samples.

Cytoplasm		Membranes	
Accession	Function	Accession	Function
FGSG_00421	S-Adenosylmethionine synthetase	FGSG_01962	Argininosuccinate synthase
FGSG_01346	Enolase	FGSG_02077	Hypothetical protein
FGSG_01591	Phosphate carrier protein	FGSG_02937	Hypothetical protein
FGSG_02461	Malate dehydrogenase	FGSG_05997	UDP-galactopyranose mutase
FGSG_02473	Mitochondrial ATP synthase subunit F	FGSG_06290	Serine hydroxymethyltransferase
FGSG_02523	IgE-dependant histamine-releasing factor	FGSG_06384	COPII-coated vesicle component SEC24
FGSG_03569	Galactose oxidase precursor	FGSG_07457	rRNA processing protein EBP2
FGSG_04826	L-Xylulose reductase	FGSG_09366	Glucan 1,3- β -glucosidase
FGSG_05035	Pyridoxine biosynthesis protein	FGSG_09454	kap95 protein
FGSG_05615	Adenosyl homocysteinease	FGSG_10267	C-1-tetrahydrofolate synthase
FGSG_06246	Heat-shock protein 60	FGSG_10974	Hypothetical protein
FGSG_08737	Woronin body major protein	FGSG_16615	Hypothetical protein
FGSG_09896	Isocitrate lyase	FGSG_16707	Hypothetical protein
FGSG_10549	Hypothetical protein		
FGSG_10746	Aspartate aminotransferase		
FGSG_10942	Hypothetical protein		
FGSG_11627	HTA2-histone H2A.2		
FGSG_12851	Allantoicase		
FGSG_17593	Translation activator GCN1		

Cytoplasm and membranes	
Accession	Function
FGSG_00330	Acetyl-coenzyme A synthetase
FGSG_00496	Inorganic pyrophosphatase
FGSG_00777	Peptidyl-prolyl <i>cis/trans</i> isomerase, mitochondrial (cyclophilin B)
FGSG_00838	Heat-shock protein 70
FGSG_00921	Heat-shock protein SSB1
FGSG_00979	Aldehyde dehydrogenase
FGSG_01111	6-Phosphogluconate dehydrogenase
FGSG_01422	Citrate synthase
FGSG_01425	Plasma membrane ATPase
FGSG_01433	Glutamate synthase precursor
FGSG_04181	Elongation factor 3
FGSG_04312	H ⁺ -transporting ATP synthase β chain
FGSG_04458	Nitric oxide dioxygenase
FGSG_04527	Carboxypeptidase S1
FGSG_05972	Nucleoside diphosphate kinase
FGSG_06021	ADP, ATP carrier protein
FGSG_06566	Succinate/fumarate transporter
FGSG_07401	Elongation factor 1- γ 1
FGSG_08601	Phosphoenolpyruvate carboxykinase
FGSG_08700	Malate synthase
FGSG_08811	Elongation factor α
FGSG_09574	Elongation factor 2
FGSG_09690	Peptidylprolyl isomerase
FGSG_09870	CPC2 protein
FGSG_09933	Mitochondrial porin
FGSG_10118	Ketol-acid reductoisomerase
FGSG_10825	5-Methyltetrahydropteroyltriglutamate-homocysteine methyltransferase
FGSG_10855	Alcohol dehydrogenase 1
FGSG_15824	ATP synthase subunit α
FGSG_16627	Glyceraldehyde-3-phosphate dehydrogenase

The distribution of proteins is shown in the Venn diagram.



the genes being involved in the disease onset process (Pan et al., 2018).

Functional characterization of CEPs by deletion and overexpression analysis

Out of 62 CEPs, 19 have been characterized previously to play a role in virulence, representing 31% (19/62) of CEPs isolated by BioID (Table 3). For this study, we chose 12 CEPs for functional characterization (Table 4). We used the RNP/CRISPR/Cas9 and homologous recombination method to delete each CEP (Abdallah et al., 2017). The deletions were confirmed by PCR and whole-genome sequencing of two mutant strains, Δ FGSG_00979 and Δ FGSG_05615, was performed to ensure that no ectopic indels were present (Figure S1). Next, we performed infection assays on wheat heads (cv. Roblin), and out of the 12 deleted CEPs, only FGSG_00979 and FGSG_05615 had reduced virulence on wheat heads (Figure 3a). FGSG_05615 encodes for a hydrolase, designated as FgSah1, and converts *S*-adenosyl-L-homocysteine (AdoHcy) to *S*-adenosyl-L-methionine (AdoMet) (Shi et al., 2021). It is highly conserved in fungi and deletion of this gene in *F. graminearum* results in the accumulation of AdoHcy (Shi et al., 2021). Similar to the findings in our study, the FgSah1 mutant showed a decrease in virulence on both flowering wheat heads and coleoptiles; however, the mutant is competent to produce more DON than the wild-type PH-1 strain (Shi et al., 2021). It should be noted that the mutant grows poorly even in a rich medium and, therefore, the disease phenotype is likely to be associated with the reduced fitness level of the mutant strain. Functional orthologs of FGSG_00979 are

present in other *Fusarium* species (Díaz-Sánchez et al., 2016). The gene encodes an aldehyde dehydrogenase capable of converting retinal into retinoic acid in *Fusarium fujikuroi* and *Fusarium verticillioides*. The mechanism of action has not been elucidated; however, the impact of light, acting specifically through carotenoids, on many developmental processes in fungi, including *F. graminearum*, suggest that FGSG_00979 will have a strong impact in the growth and development of *F. graminearum*.

As virulence factors have been shown to display different phenotypes on vegetative tissues compared with the inflorescence wheat heads, we used the same 12 deleted CEP strains and performed pathology tests on the wheat coleoptiles (cv. Roblin). This assay revealed an additional five CEPs involved in virulence: FGSG_02077, FGSG_09690, FGSG_10746, FGSG_04527 and FGSG_10549 (Figure 3b). Similar to previous studies, the mutation of FGSG_02077 reduced virulence in wheat coleoptiles (Zhang et al., 2012). FGSG_10549 is annotated as a hypothetical protein; however, with its high cysteine content, it is likely to belong in the same family of effectors as FGSG_02077. Other CEPs characterized by coleoptile assay included FGSG_09690, possessing peptidylprolyl isomerase (PPI) domains, which are found in cyclophilins. Cyclophilins catalyze peptide bonds in proline residues and facilitate protein folding (Wang & Heitman, 2005). The deletion of the cyclophilin homolog CYP1 in *Magnaporthe grisea* showed a reduction in virulence (Viaud et al., 2002). In the parasite *Trypanosoma cruzi*, an FK506 binding protein with a PPI domain was shown to be secreted and involved in cell infection (Moro et al., 1995).

Table 3 Candidate effector proteins identified by BioID with function in fungi

	CEP	Annotation	Reference
1	FGSG_10825	Homocysteine methyltransferase	Kim et al. (2015)
2	FGSG_00838	Hsp 70	Liu et al. (2017)
3	FGSG_05615	<i>S</i> -Adenosyl-L-homocysteine hydrolase	Shi et al. (2021); This study
4	FGSG_02077	CFEM1	Zhang et al. (2012); This study
5	FGSG_04181	ABC transporter	Yin et al. (2018)
6	FGSG_08737	Woronin body protein	Fernando et al. (2019)
7	FGSG_10118	Keto-acid reductoisomerase	Liu et al. (2014)
8	FGSG_00921	Hsp 70	Liu et al. (2017)
9	FGSG_09933	Mitochondrial porin	Maeda and Ohsato (2017)
10	FGSG_04458	NS response gene <i>FgFHB2</i>	Jian et al. (2021)
11	FGSG_09896	Isocitrate lyase 1	Kim et al. (2015)
12	FGSG_16627	Glyceraldehyde 3-phosphate DHG	Wang et al. (2018)
13	FGSG_17593	Translational activator Gcn1	Wang et al. (2018)
14	FGSG_15824	ATP synthase subunit – mitochondrial	Wang et al. (2018)
15	FGSG_03569	Galactose oxidase precursor	Zheng et al. (2018)
16	FGSG_10974	Hypothetical protein	Zheng et al. (2018)
17	FGSG_09690	Peptidyl-prolyl <i>cis/trans</i> isomerase	Liu et al. (2010); This study
18	FGSG_09870	G protein in <i>Fusarium verticillioides</i>	Yan and Shim (2019)
19	FGSG_01425	H ⁺ -ATPase	Wu et al. (2021)

CEP, candidate effector protein.

Table 4 Phenotypes of the 12 candidate effector proteins characterized in this study

	CEP	Rank ^a	Annotation	Phenotype ^b
1	FGSG_09366	5	Glucan 1,3-β-glucosidase	No
2	FGSG_04826	5	L-Xylulose reductase	No
3	FGSG_00979	4	Aldehyde dehydrogenase	Yes
4	FGSG_05615	4	S-Adenosyl-L-homocysteine hydrolase	Yes
5	FGSG_09690	3	Peptidyl-prolyl <i>cis/trans</i> isomerase	Yes
6	FGSG_02077	3	CFEM1	Yes
7	FGSG_01422	3	Citrate synthase	No
8	FGSG_10974	3	Hypothetical protein	No
9	FGSG_10746	3	Aspartate aminotransferase	Yes
10	FGSG_10855	3	Alcohol dehydrogenase 1	No
11	FGSG_04527	2	Carboxypeptidase S1	Yes
12	FGSG_10549	1	Hypothetical protein	Yes

CEP, candidate effector protein.

^aRank is based on the presence of five data sets, shown in Table S4.

^bPhenotype assessed in either wheat inflorescence or vegetative tissues.

Nutrient acquisition by pathogens is critical for the proliferation and colonization of host cells, and in this context metabolic enzymes play an important role. Several metabolic enzymes were identified by BioID that function as virulence factors. Aspartate aminotransferase (AAT) is important in nitrogen utilization and is critical for pathogenicity in the oomycete *Phytophthora sojae* (Wang et al., 2016). The deletion of AAT (encoded by FGSG_10746) in this study also showed that it has a role in virulence in *F. graminearum* (Figure 3b).

To ensure that the phenotypes observed with the deletion mutant strains are not caused by fitness defects affected by the mutation, we performed a growth curve of the deletion strains in a liquid growth medium (potato dextrose broth). As shown, except for FGSG_05615, which has previously been shown to have a fitness defect, all other deletion mutants grew similarly to the wild-type *F. graminearum* strain (Figure S2; Shi et al., 2021).

Both the wheat head and the coleoptile pathology tests provided us with information about the virulence of CEPs, but we were also interested to know how the CEPs might affect plant immunity in the context of the vast effector repertoire present in *F. graminearum*. Towards this goal, we adapted a pathogenicity test that involved the constitutive expression of the seven CEPs that were confirmed to possess virulence function (Figure 3; Table 4) and assessed the growth of the virulent bacterial pathogen *P. syringae* DC3000 in wheat (cv. Roblin). This bacterial strain grows in wheat and induces necrosis (Figure S3) (Ramachandran et al., 2017). The overlap pathology assay is quantitative, and therefore an increase or decrease of bacterial growth compared with the control (extracellular vesicle, EV; Figure 4a) indicated whether a given effector candidate suppresses or activates plant immunity. Various iterations of the overlap assay have been shown to function effectively to assess plant immunity. For example, in

N. benthamiana, the non-pathogen *P. fluorescens* induced immunity and protected the plant against both the defense and the disease-triggered cell death caused by the *P. syringae* pathogens (Chakravarthy et al., 2010). In another example, the function of the pathogenesis-related protein 1 (PR1) from wheat was shown to be modulated by the fungal effector SnTox3 (Sung et al., 2021). The wheat leaves infiltrated with PR1 induced plant defense and reduced disease symptoms; however, upon an interaction between the effector SnTox3 and PR, the defense was compromised (Sung et al., 2021).

To address the role of CEPs in plant immunity, the seven CEPs that displayed phenotypes in both the wheat heads and the coleoptiles were constitutively expressed and delivered into the wheat coleoptiles through the *U. hordei* delivery system (Alonso et al., 2020). The expression of CEPs in *Ustilago* before delivery was confirmed by Western blot analysis (Figure S4). The constitutive overexpression of all seven CEPs in the wheat leaves (cv. 'Roblin') reduced the growth of *P. syringae* DC3000 in the overlap region (Figure 4). The growth was assessed against the growth of *Ustilago* containing an empty vector, and thus enhanced resistance is likely to result from the effect of the expression of the specific CEP and not be the result of any priming of the immune system by *Ustilago*. The wheat cultivar 'Roblin' is 'insensitive' to SnTox3 and its expression results in reduced *P. syringae* DC3000 growth, and therefore it was used as a positive control in our assay (Figure S5) (Hafez et al., 2020). It should be noted that co-incubation of *Ustilago* expressing the CEPs did not affect the growth of DC3000 (data not shown). Moreover, neither the expression of EV nor another candidate effector FGSG_01422 delivered by *Ustilago* affected DC3000 growth *in planta* (Figure 4; Figure S3). Although highly ranked, the two CEPs FGSG_09366 and FGSG_04826 did not display any virulence function in any of the three

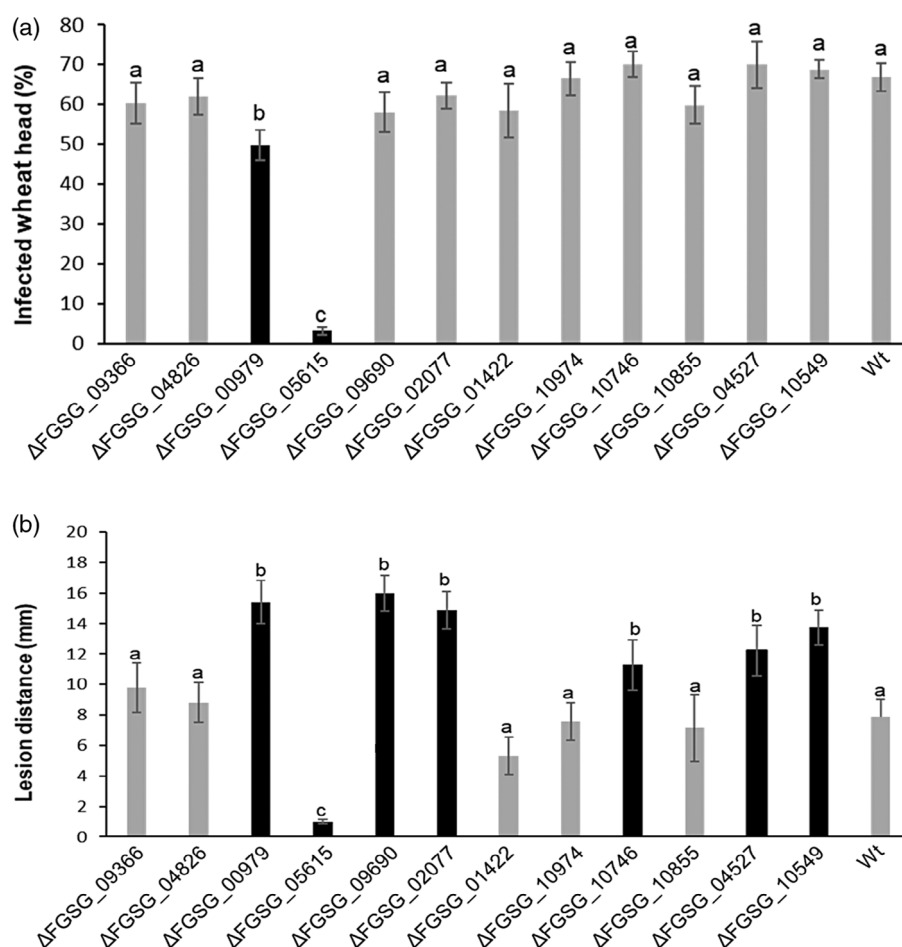


Figure 3. Pathology experiments with mutant candidate effector protein (CEP) strains performed on the inflorescence (flowering wheat heads) and vegetative (coleoptiles) tissues.

(a) Twelve mutant CEP strains were assessed for virulence on flowering wheat heads ($n = 30$) of the susceptible wheat variety (cv. 'Roblin'). The wheat spikelets were point-inoculated with 10 μ l (1000 spores) of mutant CEP *Fusarium graminearum* strains (x-axis) and the spread of the infection (%) (y-axis) was evaluated after 14 days. Each CEP was ranked based on its presence in five data sets.

(b) The same 12 mutant CEP strains were assessed for virulence on wheat coleoptiles (cv. 'Roblin'). The spores from each mutant strain (x-axis) were inoculated with 1 μ l of *F. graminearum* spores (4×10^6 ml $^{-1}$) directly applied to the cut coleoptile tips. Infection was monitored 4 days post-inoculation by measuring lesion distance (y-axis) and compared with water-inoculated coleoptiles. The dark bars indicate strains with significant differences between the wild type (Wt) and other mutant strains. Statistical analysis was performed by one-way analysis of variance (ANOVA) with Tukey's test. Letters on top of the bars represent significant differences between the stimulus and the Wt at $P < 0.05$. All experiments were repeated three times with similar results.

assays to monitor effector function. Collectively, the constitutive expression of a subset of CEPs triggered an immune response in wheat, confirming our hypothesis that these candidate secreted proteins possess effector function.

To validate our observations of immune response activation, we monitored the induction of the defense response gene *PR1* in wheat leaves inoculated with *Ustilago* expressing CEPs (Figure 4b). Compared with the EV control, the delivery of a subset of CEPs – FGSG_05615, FGSG_02077, FGSG_10746 and FGSG_04527 – induced *PR1* expression (Figure 4b). As the *PR1* expression was monitored after 2 days of CEP infiltration, combined with the observation that no *PR1* expression was observed by *Ustilago* alone (EV), we suggest that the defense mounted by

the CEPs is the result of effector-triggered immunity (Kanja & Hammond-Kosack, 2020).

Currently, no temporal or spatial patterns of effectors have been demonstrated in *F. graminearum*, but the 62 CEPs identified in this study have provided a road map to further functional studies of effectors. The characterization of effectors is an ongoing challenge; however, machine learning tools are being advanced to construct a pangenomic effectome for *Fusarium* that will considerably improve the annotation of effectors, assign effector families and reveal their function in fungi (Alouane et al., 2021). Analysis of 62 CEPs with SIGNALP indicated only two CEPs with signal peptides; however, DEEPLOC indicated that the majority of the CEPs are localized to several organelles (Table S4). Much remains to be learned

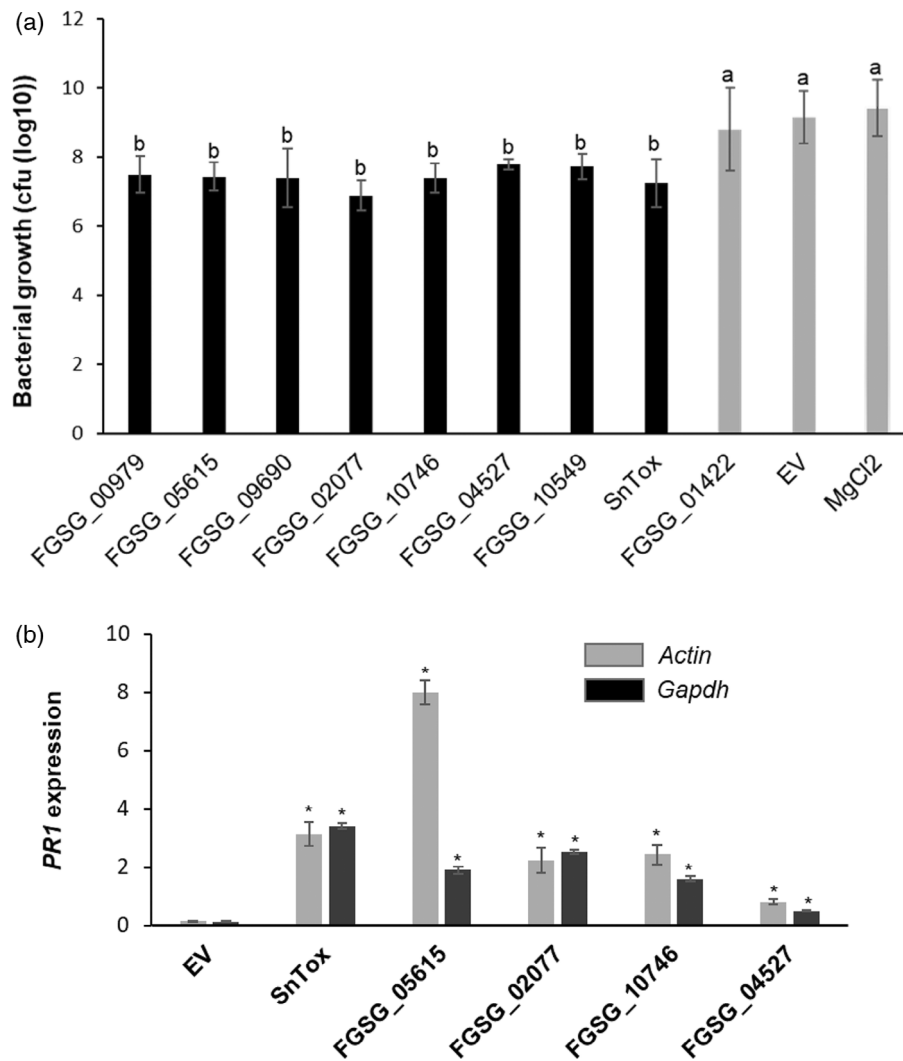


Figure 4. Candidate effector proteins (CEPs) modulate plant defense.

(a) The CEPs constitutively expressed in *Ustilago hordei* were infiltrated into 3-week-old wheat (cv. 'Roblin') leaves (two leaves per plant for 15 plants). After 2 days, the area infiltrated with *U. hordei* expressing CEPs was inoculated with the *Pseudomonas syringae* DC3000 bacteria strain. After incubation for three additional days, the overlap region was cut, weighed, homogenized, serially diluted and plated to count colony-forming units (CFUs). The numbers represent four technical replicates. SnTox3 was used as a positive control, FGSG_01422 was used as a negative control, and *U. hordei* with empty vector (EV) was used as a reference for statistical analysis. All the statistical analysis was performed by one-way analysis of variance (ANOVA) with Tukey's test. Letters on top of the bars represent significant differences between the EV and the various *Fusarium graminearum* mutant strains at $P < 0.05$. Experiments were repeated at least two times with similar results.

(b) Defense gene *PR1* induction in wheat leaves inoculated with *Ustilago* expressing CEPs normalized to two housekeeping genes: *Actin* (light-gray bars) and *Gapdh* (black bars). *Significant difference from the EV control using the Student's two-tailed t -test ($P < 0.05$) with four technical replicates. Similar trends were observed across two independent experiments.

about how *Fusarium* effectors are translocated. A recent study that examined the cargo of EVs in *F. graminearum* suggests a mechanism by which effectors might enter host cells (Garcia-Ceron et al., 2021). The next step would be to isolate and examine the EVs from infected tissues. Nevertheless, a comparison of the content of the EVs with the CEPs from this study showed an approximately 30% overlap, offering confidence in the BioID technology, and new iterations of BioID, such as TurboID, hold considerable potential to both characterize additional effectors and identify their interactors in

plants. Finally, the simultaneous deletion of multiple effectors by the CRISPR/Cas9 technology and localization of effectors with the split GFP technique will also advance our understanding of effector function (Park et al., 2017).

EXPERIMENTAL PROCEDURES

Arabidopsis–*F. graminearum* infection assay and imaging

Transgenic Arabidopsis (Col-0) expressing cytoplasmic BirA* (cBirA*) and membrane BirA* (mBirA*) were used in

Arabidopsis–*F. graminearum* infection assays (Khan et al., 2018). Briefly, between five and seven Col-0, *BirA** or *HopF2-BirA** Arabidopsis seeds were sterilized and placed in each well of a 96-well tissue culture plate (Sarstedt, <https://www.sarstedt.com>). A 200- μ l aliquot of 0.5 $\times$ MS (Murashige and Skoog basal media, 2.5 mM 2-(*N*-morpholino) ethanesulfonic acid, MES) was added to each well and stratified in the dark for 2 days at 4°C on a Heidolph Titramax 1000 vibrating shaker. Plates were then moved into a growth chamber at 22°C under long days (16 h light/8 h dark) for 5 days, at which point the liquid media was removed and 200 μ l of *F. graminearum* (NRRL29169) spores were added (800 spores ml⁻¹ in 0.5 $\times$ MS) (Schreiber et al., 2011). Plates were incubated for 24, 48 or 60 h, with gentle shaking. D-Biotin (1 mM; BioBasic, <https://www.biobasic.com>) and 30 μ M of dexamethasone (Dex; Sigma-Aldrich) were added to each well 24 hours before collecting seedlings. Lactophenol trypan blue staining was used to visualize cell death and fungal growth on infected seedlings (Koch & Slusarenko, 1990).

Optimization of biotin levels for infection assay and confirmation of *BirA** expression by immunoblot analysis

To optimize biotin levels, concentrations ranging from 0 to 2 mM of biotin were added to seedlings and *BirA** was induced with 30 μ M Dex. At 24 h after the addition of biotin and Dex, seedlings from each biotin concentration were pooled, ground to a powder with liquid nitrogen and suspended in protein extraction buffer (20 mM Tris, pH 8.0, 100 mM NaCl, 0.4% SDS, 0.1% Triton X-100, 1 mM DTT, and protease inhibitor cocktail; Sigma-Aldrich, <https://www.sigmaaldrich.com>). Samples were incubated on ice for 30–40 min with occasional vortexing, then spun down at 16 000 *g* for 15 min at 4°C. The protein supernatant was quantified with Fourier Transform Infrared (FTIR) spectroscopy (Direct Detect Infrared Spectrometer; Merck, <https://www.merckmillipore.com>). An aliquot (20–30 μ g) of protein extract in Laemmli buffer was separated by a 12% SDS-PAGE. Proteins were transferred onto a nitrocellulose membrane, blocked overnight in 3% ECL blocking buffer (Amersham BioSciences, now GE Healthcare, <https://www.gehealthcare.com>), and then incubated for 1 h with either streptavidin-HRP conjugate (GE Healthcare) or HA antibodies (Roche, <https://www.roche.com>). Membranes were developed with Lumigen ECL Ultra solutions (Lumigen, <http://www.lumigen.com>) and imaged with the iBright750 imaging system (Invitrogen, now ThermoFisher Scientific, <https://www.thermofisher.com>).

Affinity purification of biotinylated proteins

Three biological replicates were used for affinity purification and mass spectrometry analysis. Each replicate with approximately 400 mg of Arabidopsis was used for protein extraction, as described previously. An aliquot of the supernatant was used as the input fraction for immunoblot analysis. Proteins were concentrated and excess biotin was removed by centrifugation with an Amicon Ultra-4 (Ultracel-10) filter column (Millipore, now Merck) at 3000 *g* for 5 min. The filter columns were rinsed with 3 ml of extraction buffer (with no detergents) and centrifuged at 4000 *g* for 15 min. The rinse step was repeated until the extract was concentrated to a final volume of 500 μ l.

Approximately 1 mg of protein was incubated with 50 μ l pre-washed streptavidin-Sepharose high-performance beads (GE Healthcare) for 1 h at 4°C. Beads were spun down for 2 min at 500 *g* and the supernatant (unbound fraction) was saved for immunoblot analysis. Beads were rinsed once with 2% SDS, 200 mM Tris, pH 8.0, and three times with 20 mM Tris, pH 8.0, and 100 mM NaCl. A 50- μ l aliquot of resuspended beads was set aside

as the bound fraction for immunoblot analysis. After the final rinse, all supernatant was removed and the dry beads were sent for mass spectrometry. Immunoblot analyses were performed as before using input, bound and unbound fractions. A 2-mm aliquot of biotin was added to the bound fraction before boiling to prevent the rebinding of protein to the beads.

Isolation of secreted proteins from *F. graminearum* axenic cultures

Fusarium graminearum (NRRL29169) was grown in the nutrient-rich medium at 5000 spores ml⁻¹ for 24 h (Rampitsch et al., 2010). The samples were then transferred to the nutrient-limiting conditions in which a surface-sterilized wheat head (*T. aestivum*, mid-anthesis) was added to each sample (Rampitsch et al., 2010). Samples were incubated further for 24 h. Culture supernatants were filtered and proteins were precipitated overnight using a TCA–acetone–DTT procedure (Rampitsch et al., 2010). Isolated proteins were dried over N₂ gas, digested with trypsin, using standard procedures, and subjected to liquid chromatography with tandem mass spectrometry (LC-MS/MS) analysis using the same mass spectrometry approach and search strategy as described below.

LC-MS/MS analysis to identify *F. graminearum* peptides

Dried beads with affinity-purified proteins were digested on bead with trypsin as described by Hesketh et al. (2017). A description of each treatment or control sent for mass spectrometry analysis is described in Table 2. Tryptic peptides were separated through a C18 column (12-cm fused silica column, 75- μ m inner diameter, packed with Vydac C18, 5- μ m beads, 300 Å pores) coupled to an Orbitrap Q Exactive mass spectrometer (ThermoFisher Scientific) via a nanoelectrospray ionization source. An acetonitrile gradient of 2%–50% (v/v) acetonitrile in 0.1% (v/v) formic acid was delivered at 300 nl min⁻¹ over 60 min, with a total run of 120 min. A survey scan was acquired over an *m/z* range of 300–2000, followed by 12 MS/MS scans of the most intense ions. To prevent cross-contamination, control samples were injected first into the mass spectrometer. MASCOT 2.5.1 (MatrixScience, <https://www.matrixscience.com>) was used for protein identification. Raw mass spectrometry data files were converted to MGF using MASCOT DISTILLER and queried against the SwissProt Arabidopsis database (<https://www.uniprot.org>) (535 248 sequences) and the MIPS *F. graminearum* database from PH-1 and the strain NRRL29169 (Güldener et al., 2006; Rampitsch et al., 2013). MASCOT searches used to identify proteins were set with the following parameters: monoisotopic mass accuracy of ± 5 ppm, up to two missed cleavages, peptide charge up to +5, fixed modification of carbamidomethyl and variable modifications of oxidation (M) and biotin (K, N-term). The false-discovery rate (FDR) was set to 1% and validated against a target–decoy database. Proteins were considered correctly identified if hits contained two or more peptides with a significant ion score, based on the probability of the match being a random event (Perkins et al., 1999). MASCOT uses a hierarchical clustering algorithm to group proteins with shared peptides into families and these are reported together, along with any unique peptides observed. To manage the misidentification of proteins when spectra from unique, proteotypic peptides were not acquired during mass spectrometry, a manual inspection was used and verified that the MASCOT algorithm was returning correctly matched proteins (Carr et al., 2004). The RAW spectrum files have been deposited in PRIDE (<https://www.ebi.ac.uk/pride>) with project identifier PXD034425. Reviewers can review the data with the username 'reviewer_pxd034425@ebi.ac.uk' and password 'l5ZiqWKK'.

Identification of candidate *F. graminearum* effector proteins

Fusarium graminearum proteins identified with MASCOT searches were filtered from Arabidopsis proteins and a series of filters were applied to generate a list of CEPs. Each protein identified by MASCOT was assigned a protein score (derived from the ion score) to represent significant (95% confidence) identity and homology to the reference protein based on the size of the reference database and the number of peptides found from mass spectrometry analysis. When using the *F. graminearum* MIPS database, peptides with a score above 20 were considered significant (95% confidence) with a strong identity and homology to the matching protein. *Fusarium graminearum* proteins with a score above 20, present in two of three biological replicates and absent in any control treatment were considered to be reliably identified *F. graminearum* proteins. The data sets are presented in Datasets 1–5.

Analysis of CEPs: software and published data sets

Secretion signals and localization for CEPs were assessed based on SIGNALP 5.0 scores and DEEPLOC 2.0, respectively (Almagro Armenteros et al., 2017; Almagro Armenteros et al., 2019; Emanuelsson et al., 2000). CEPs were also analyzed with EFFECTORP 1.0 and EFFECTORP 2.0 (Sperschneider et al., 2016, 2018). The number of cysteine residues was determined with EXPASY PROTPARAM (Gasteiger et al., 2005). Functional enrichment analysis of CEPs was performed by FUNGIFUN 2 (Priebe et al., 2015). Protein domains were identified with PFAM 32.0 from EMBL-EBI (<https://www.ebi.ac.uk>) (El-Gebali et al., 2019). CEPs were mapped to the *F. graminearum* genome (from MIPS) using FGRA3MAP and aligned with regions of high recombination and high single-nucleotide polymorphism (SNP) density (Antoniw et al., 2011). CEPs were ranked based on their presence in the published secretome, proteome and transcriptome data sets (where a score of 5 = present in five data sets) (Fabre et al., 2019; Pan et al., 2018; Paper et al., 2007; Rampitsch et al., 2013).

Gene deletion and gene insertion by CRISPR/Cas9

Two CRISPR RNAs (crRNAs) with no off-target or partial targets were designed for each gene of interest using EuPaGDT (<http://grna.ctegd.uga.edu/>) with *F. graminearum* PH-1 database FungiDB-26 and with the SpCas9 settings: 20 nt guide RNA (gRNA) and NGG protospacer-adjacent motif (PAM) on 3' end. All crRNA, trans-activating crRNA (tracrRNA) and Cas9 were purchased from IDT (<https://eu.idtdna.com>) (Figure S1). The repair template along with the gRNA complex was assembled with 5 µl each of the crRNA (100 µM stock), the tracrRNA (100 µM stock) and the nuclease-free duplex buffer (IDT) to a final concentration of 33 µM in 15 µl. The ribonucleoprotein (RNP) complex was assembled by the addition of 1.5 µl of 33 µM gRNA, 0.75 µl of 1 µg ml⁻¹ Cas9 and 11 µl of Cas9 working buffer (200 mM HEPES, 150 mM KCl, pH 7.5), incubated for 10 min at 22°C and transferred into *F. graminearum* protoplasts. A full list of primers can be found in Table S5.

Preparation of *F. graminearum* protoplasts: fresh fungal spores were inoculated into 100 ml of YEPD medium (3 g yeast extract, 10 g peptone and 20 g dextrose per 1 L water) at a concentration of 1×10^8 ml⁻¹. Spores were germinated until the germ tube is approximately 1.5–2.0 times the length of the conidium (approx. 8–10 h at 28°C and gentle shaking). Germ tubes were harvested through filtration with twofold sterile Miracloth (Millipore, now Merck), washed once with water and once with 30 ml

of 1.2 M KCl. Washed mycelia were then transferred to the protoplast solution (500 mg Driselase, 200 mg Yatalase and 200 mg Lysing enzyme per 20 ml of 1.2 M KCl, sterile-filtered at 0.45 µm) for 30 min at 28°C with gentle shaking. Protoplasts were filtered through a 0.4-µm cell strainer (BioBasic) and washed with 2×20 ml of 1.2 M KCl following centrifugation (2500 g for 10 min at 4°C). Protoplasts were resuspended to a final concentration of approx. 10^7 – 10^8 ml⁻¹ in STC buffer (1.2 M sorbitol, 10 mM Tris-HCl, pH 8.0, 50 mM CaCl₂) with 7% DMSO (v/v). Protoplasts were frozen at –80°C until use.

The fungal transformation was achieved by combining the entire RNP complex with 200 µl of protoplasts (10^7 – 10^8 ml⁻¹), 7 µg of PCR-purified repair template and 25 µl of a PEG-CaCl₂ buffer (60% v/w PEG-4000, 50 mM CaCl₂·H₂O, 450 mM Tris-HCl, pH 7.5) on ice for 1 h. A 1.5-ml volume of PEG-CaCl₂ buffer was then added, followed by a 20-min incubation at room temperature. The total volume was transferred to a sterile 15-ml culture tube (Falcon; Corning, <https://www.corning.com>) and made up to 2 ml with STC buffer and 3 ml of TB3 media (3 g L⁻¹ yeast extract, 3 g L⁻¹ casamino acids, 20% w/v sucrose). The mixture was then incubated for 18–20 h at 25°C. The regenerated protoplasts (300 µl) were then mixed in 20 ml of molten TB3 (0.7% low melting point agarose, 45°C) containing 100 µg ml⁻¹ hygromycin B and plated. Cultures were incubated at 28°C for 5–6 days until resistant colonies were observed on the surface of the agarose. Putative transformants were individually transferred to potato dextrose agar (PDA) with 100 µg ml⁻¹ hygromycin B for 2 days at 28°C. Single spores from transformants were propagated on PDA with 150 µg ml⁻¹ hygromycin B.

Pathology tests on wheat spikelets and coleoptiles

Triticum aestivum L. 'Roblin' seeds were surface-sterilized with bleach (50% v/v), planted and the infection of wheat spikelets was performed (Schreiber et al., 2011). For the coleoptile assays, sterilized seeds were germinated under a long-day cycle of 16-h days (20°C)/8-h nights (18°C) in a growth chamber (Sanyo, <https://www.sanyo-av.com>) with 90% relative humidity. Coleoptiles were grown for 4 days with the emergence of the first leaf before being cut back approximately 5 mm below the first leaf. A 1-µl volume of fungal spores (4×10^6 ml⁻¹) was directly applied to the coleoptile tips and the infection was then monitored, observing lesion length and browning resulting from infection at 4 days post-inoculation compared with mock treatment with water.

Overlap pathology assay in wheat by constitutive expression of CEPs

The CEPs were cloned into pENTR/D-TOPO (Invitrogen, now ThermoFisher Scientific, <https://www.thermofisher.com>), verified by sequencing and transferred into pUHS destination vector by Gateway LR Clonase II reaction (Invitrogen, now ThermoFisher Scientific) (Alonso et al., 2020). Each CEP-pUHS was transformed into *U. hordei* and cultured to an OD₆₀₀ of 1.5 (Tsukuda et al., 1988). The 3-week-old wheat coleoptiles were infiltrated into two leaves per plant (15 plants) using a 1-ml needleless syringe after puncturing the leaf once with a syringe needle (Figure S4). An approximately 2–3 cm area of wheat leaves was infiltrated, and the boundary of infiltration was marked (day 1). After 2 days, *P. syringae* pv. *tomato* DC3000 (OD₆₀₀ of 0.2) was inoculated into the previously infiltrated area such that half of the area overlapped with the *U. hordei* infiltration area (day 3). Half of the infiltrated plants were allowed to recover for approximately 1 h before the overlap area was harvested (day 3; 0 dpi). The remaining plants were incubated for an additional 3 days and harvested for

measuring bacterial growth (day 6; 3 dpi). Harvested leaf tissues were divided into four technical replicates and weighed before homogenization in 500 µl of 10 mM MgCl₂. Serial dilutions were plated (5 µl) onto a single Kings Broth plate and colony-forming units (CFUs) were calculated per gram of leaf tissue.

RNA extraction and quantitative reverse transcriptase PCR

Three-week-old wheat plants were infiltrated with *Ustilago* containing CEPs into two leaves per plant (four plants) using a 1-ml needleless syringe after puncturing the leaf once with a syringe needle. An area of wheat leaves of approximately 2–3 cm in diameter was infiltrated, and after 2 days the infiltrated leaves were harvested and total RNA was extracted using the RNeasy Mini kit, according to the manufacturer's instructions (Qiagen, <https://www.qiagen.com>). After DNase treatment, RNA was converted into cDNA by the Applied Biosystems cDNA synthesis kit according to the manufacturer's recommendations (Applied Biosystems, now ThermoFisher Scientific). The qRT-PCR reactions (15-sec denaturation at 95°C, 15-sec annealing at 60°C and 1-min extension at 72°C, repeated for a total of 40 times) were performed in quadruplicate using Applied Biosystems Power SYBR Green kit (Life Technologies, now ThermoFisher Scientific) and the QuantStudio3 Real-Time PCR System, according to manufacturer's instructions (ThermoFisher Scientific). For relative quantification, a standard curve for each primer set was created and the wheat *Actin* and *Gapdh* were used to normalize expression (Table S5).

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CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest associated with this work.

SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article.

Dataset 1. Proteins identified in cBirA* samples that are either mock infected or infected by *Fusarium graminearum* without dexamethasone.

Dataset 2. Proteins identified in cBirA* samples that are either mock infected or infected by *Fusarium graminearum* with dexamethasone.

Dataset 3. Proteins identified in mBirA* samples that are either mock infected or infected by *Fusarium graminearum* without dexamethasone.

Dataset 4. Proteins identified in mBirA* samples that are either mock infected or infected by *Fusarium graminearum* with dexamethasone.

Dataset 5. Proteins identified from growth media.

Figure S1. Characterization of mutant CEP strains.

Figure S2. Growth curves of *Fusarium graminearum* wild type (WT) and deletion mutant strains in potato dextrose broth (PDB).

Figure S3. Optimization of *Pseudomonas syringae* pv. *tomato* DC3000 growth in wheat leaves.

Figure S4. Expression of CEPs in *Ustilago hordei*.

Figure S5. Evaluation of necrosis in *Triticum aestivum* (wheat) cultivars induced by SnTox3.

Table S1. Total *Fusarium graminearum* proteins identified from all MS treatments.

Table S2. Arabidopsis proteins enriched at 60 hpi from the cytoplasm and membrane BirA*.

Table S3. *Fusarium graminearum* secreted proteins from conditioned medium.

Table S4. Ranking of CEPs by published data sets.

Table S5. Primers used in this study.

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